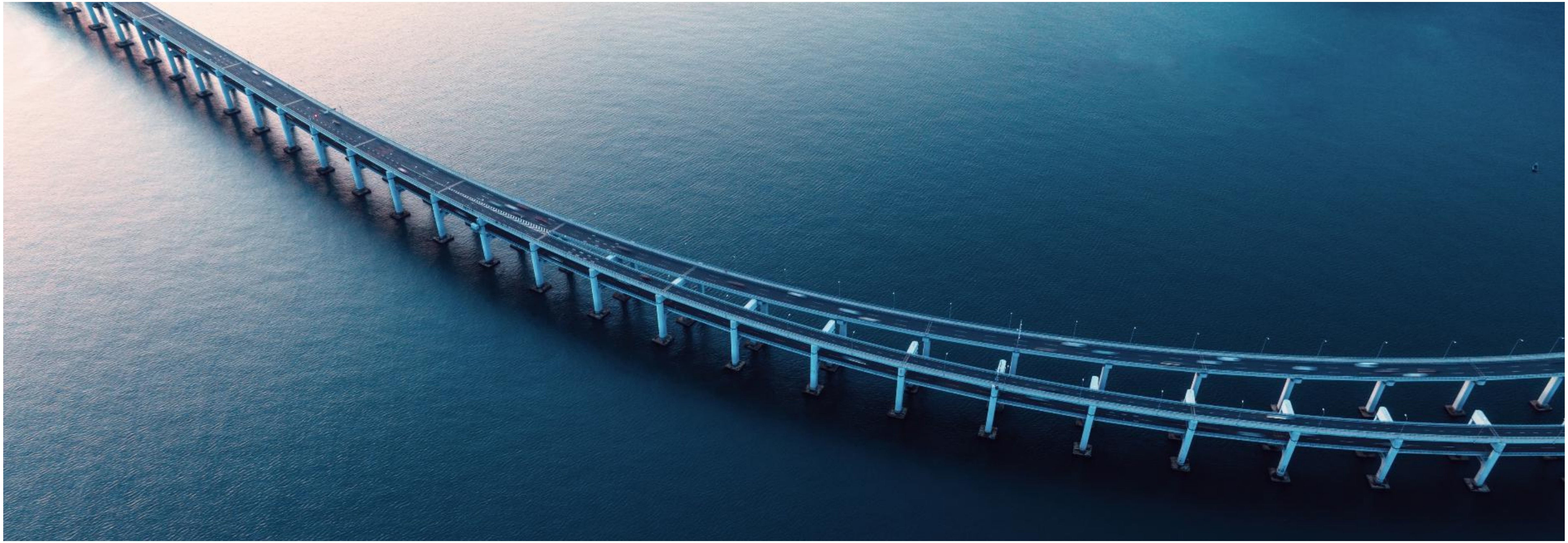
Practical Solutions Without Complexity

OpenPaths Patterns



Ben Nault, Ohio Travel Demand Model Users Group

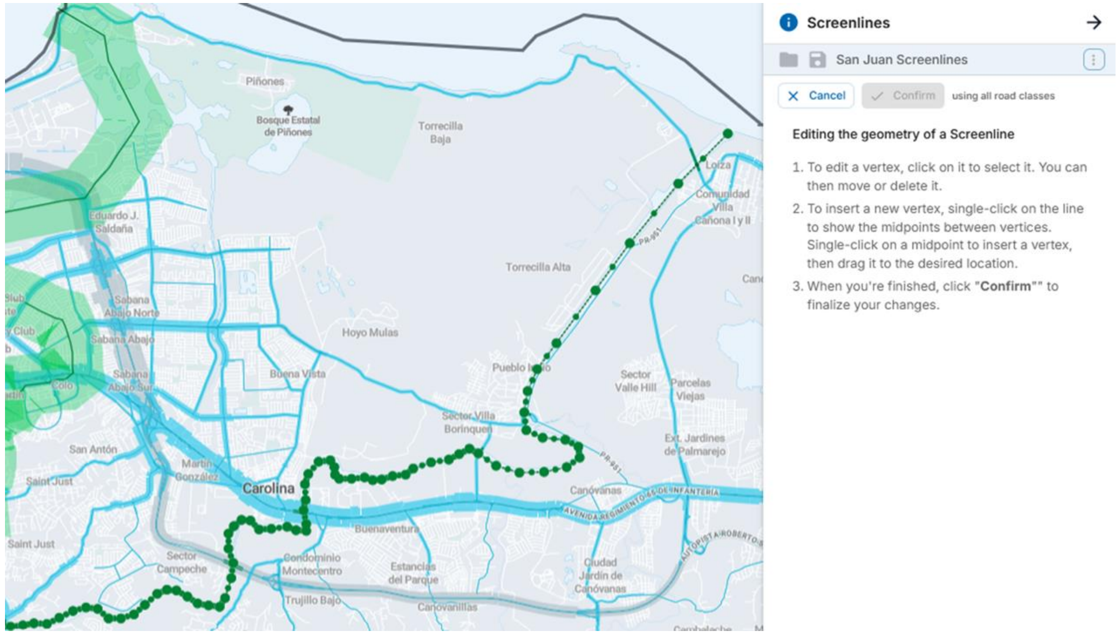
December 5th, 2025

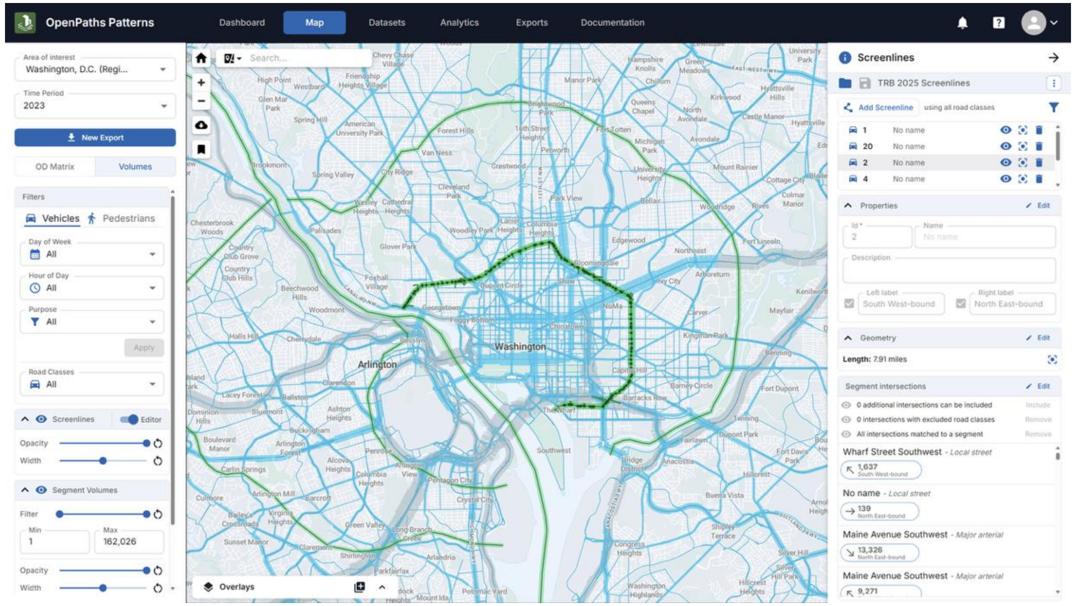
OpenPaths™ CUBE™ OpenPaths™ EMME™

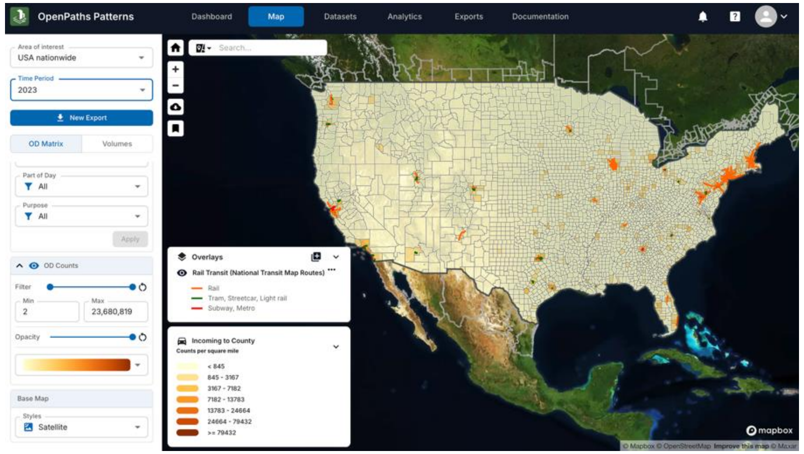
OpenPaths™ Patterns Bentley®

Agenda

- Introduce OpenPaths Patterns
- Core Features and Benefits
- Real-World Case Studies
- Q&A







OpenPaths™ Patterns

Introduction to OpenPaths Patterns

Bentley OpenPaths

Transport Planning, Modeling and Analysis

OpenPaths™

Bentley OpenPaths is used to plan, model and analyze transport systems, infrastructure and policy for more efficient, resilient, and sustainable mobility.

OpenPaths is used to build transport models, including travel demand models and traffic simulations, which provide an evidence base for strategic and operational transportation planning.

OpenPaths includes EMME, CUBE, DYNAMEQ, AGENT and CityPhi

What is OpenPaths Patterns?

OpenPaths Patterns is a web-based mobility data and analytics application for transportation analysts to harness the power of big-data for better decision-making.

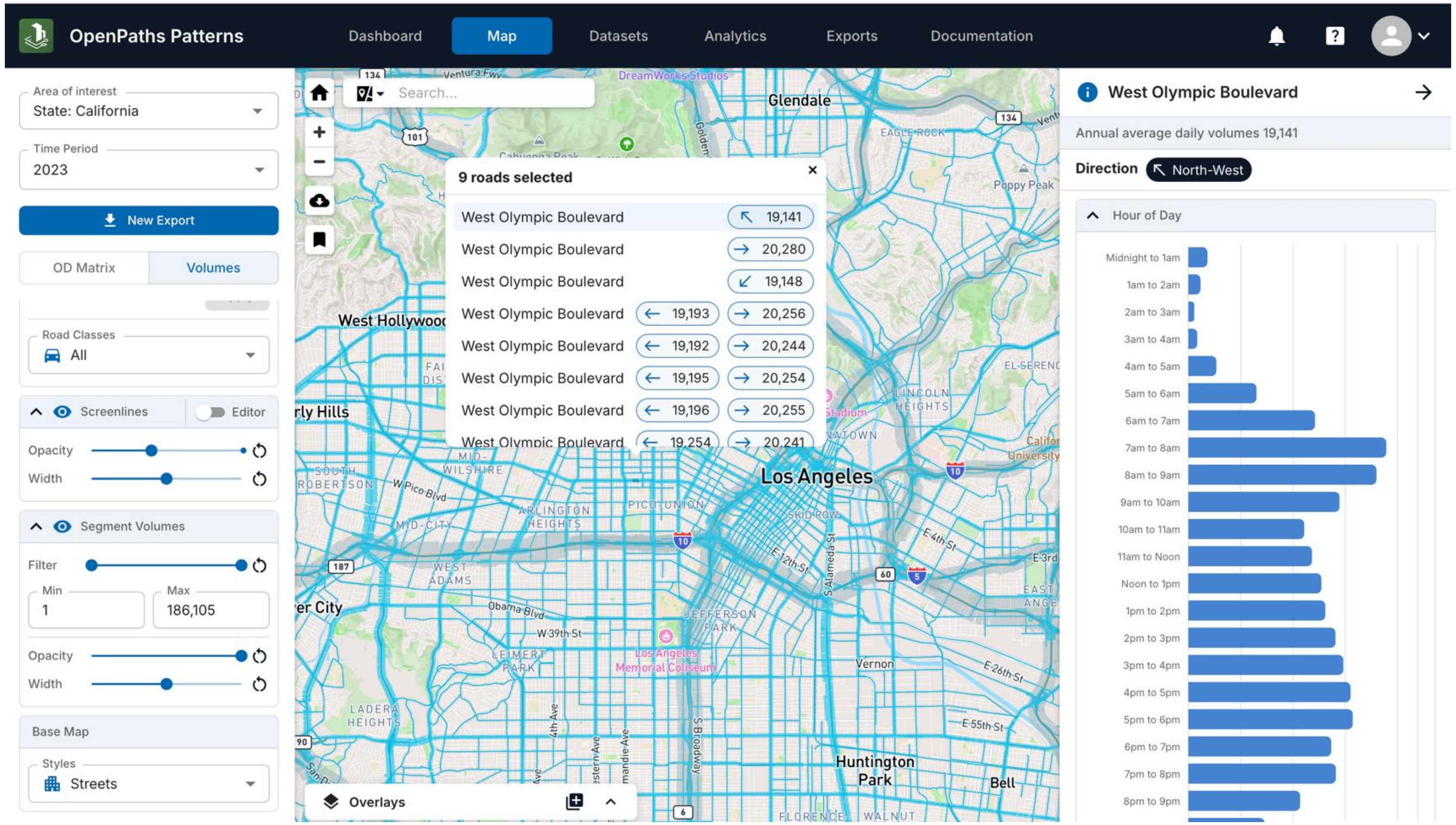
Comprehensive Mobility Data

Volume and OD Data by:

- ➢ Direction
- ➢ Hour of Day/Day Part
- ➢ Day of Week
- ➢ Trip Motivation for Driver

Supporting Travel Demand Modeling & Planning

- ✓ Safety and Risk Assessment
- ✓ Maintenance Prioritization
- ✓ Corridor Studies
- ✓ Site Location Analysis
- ✓ Advertising Insights
- ✓ Traffic Count Supplementation



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How is OpenPaths Patterns produced?

OpenPaths Patterns is built by modelers for transportation planners and modelers.

- Bentley compiles vast amounts of data to fuel a proprietary engine integrating large volumes of observed data and decades of transportation modeling research and experience into a product which creates a true representation of travel on every road in the nation.
- The result of this engine is a set of volumes on every road throughout the nation where the link on all roads is influenced by all the trips moving through the link and all ground data associated with those trips. All factors related to travel need to be considered at once, including roadway capacity, travel speed, congestion levels, geographic location, driver behavior, trip distance, trip purpose, etc. and each need to be balanced to provide a true representation of travel on the roadway.
- Three key aspects drive the process, starting with GPS-Waypoint data. The billions of data observations from the Waypoint data are converted into routed trips on the network. These routed trips are then scaled with travel behavior science and data sources such as traffic observations and land use data.

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How is OpenPaths Patterns produced?

OpenPaths Patterns is built by modelers for transportation planners and modelers.

Movement Data

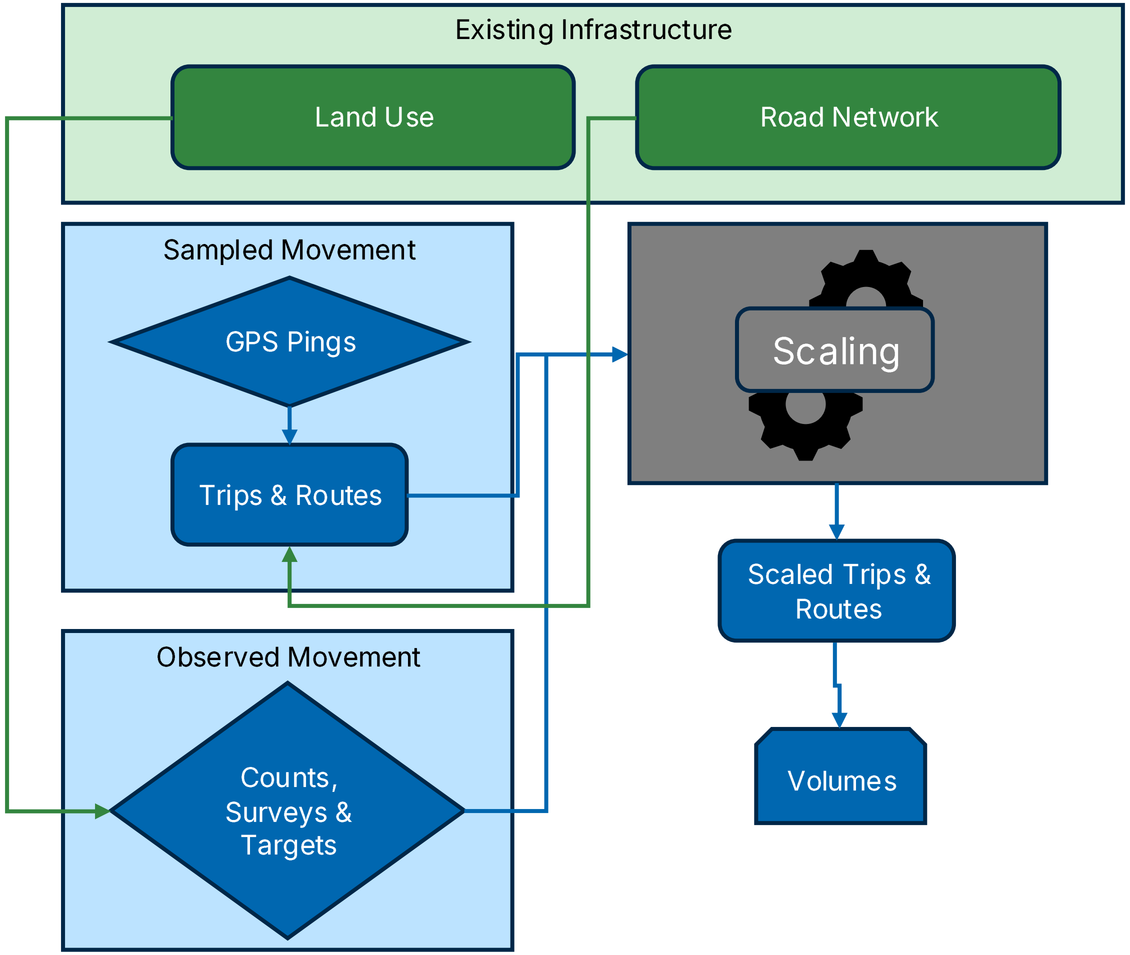
LBS data converted to trips on the network to understand how people are utilizing the system

Ground Truth

Incorporate ground truth measures, such as traffic counts, airport enplanements, employment stats

Behavior

Travel surveys feed information about why people travel



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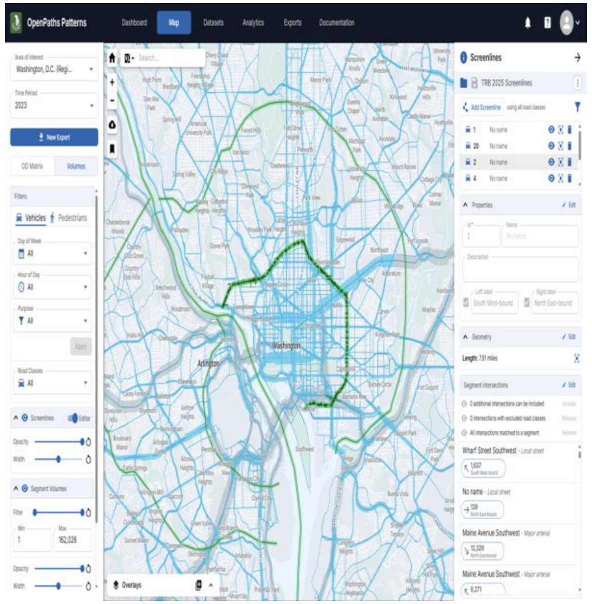
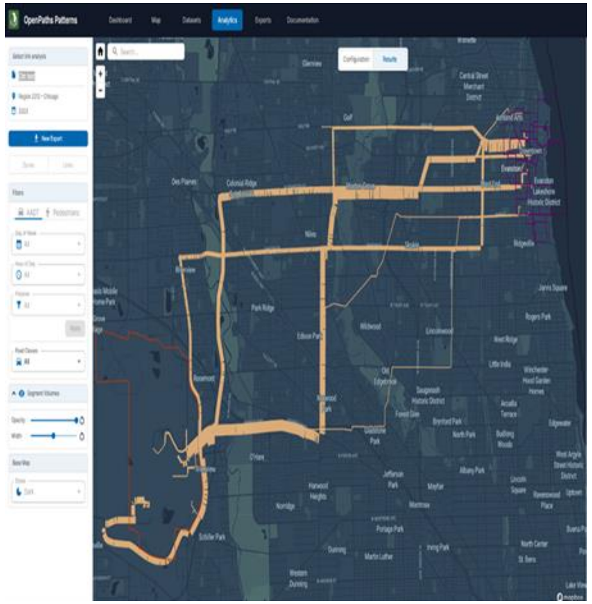
Strengths of OpenPaths Patterns

Travel Demand Modeling



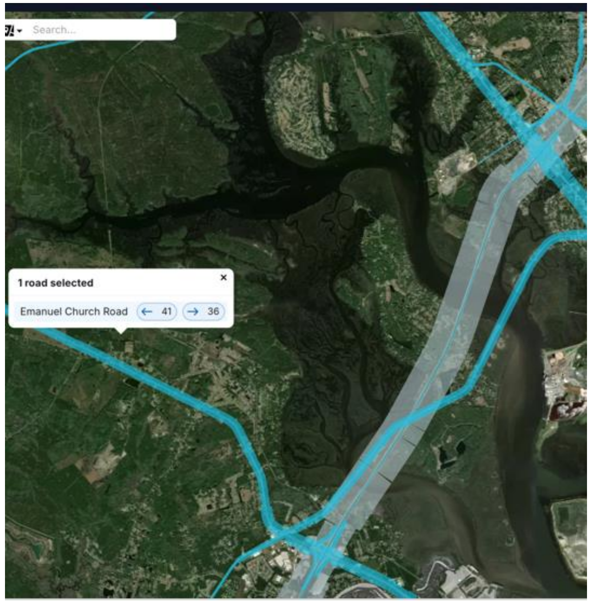
Understanding travel behavior around:
- Neighborhoods
- Recreation centers
- Hospitals, etc.

Fast Analysis



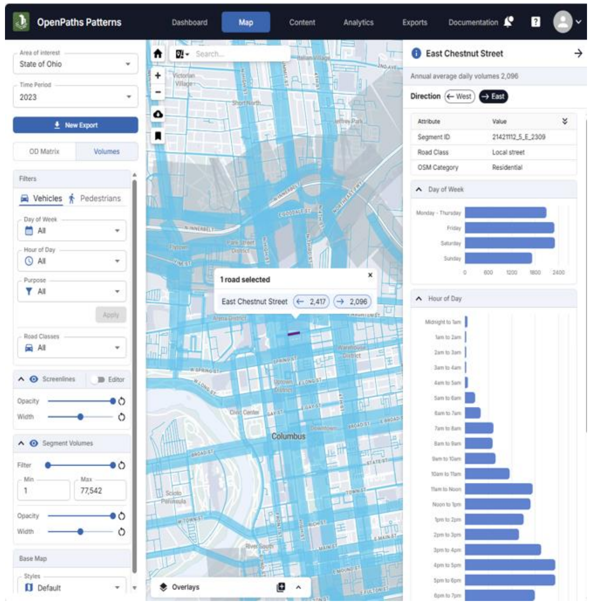
How trips interact with specific road segments and/or zones
- Site impact analysis
- Rapid identification of alternative routes
- Cut-Through traffic studies

Safety



Understand AADT on all roads
- Site impact analysis
- Rapid identification of alternative routes
- Cut-Through traffic studies

Traffic Counts



How travel flows through corridors, regions and activity centers
- Review volumes by day type, hour, trip purpose and road class combinations.

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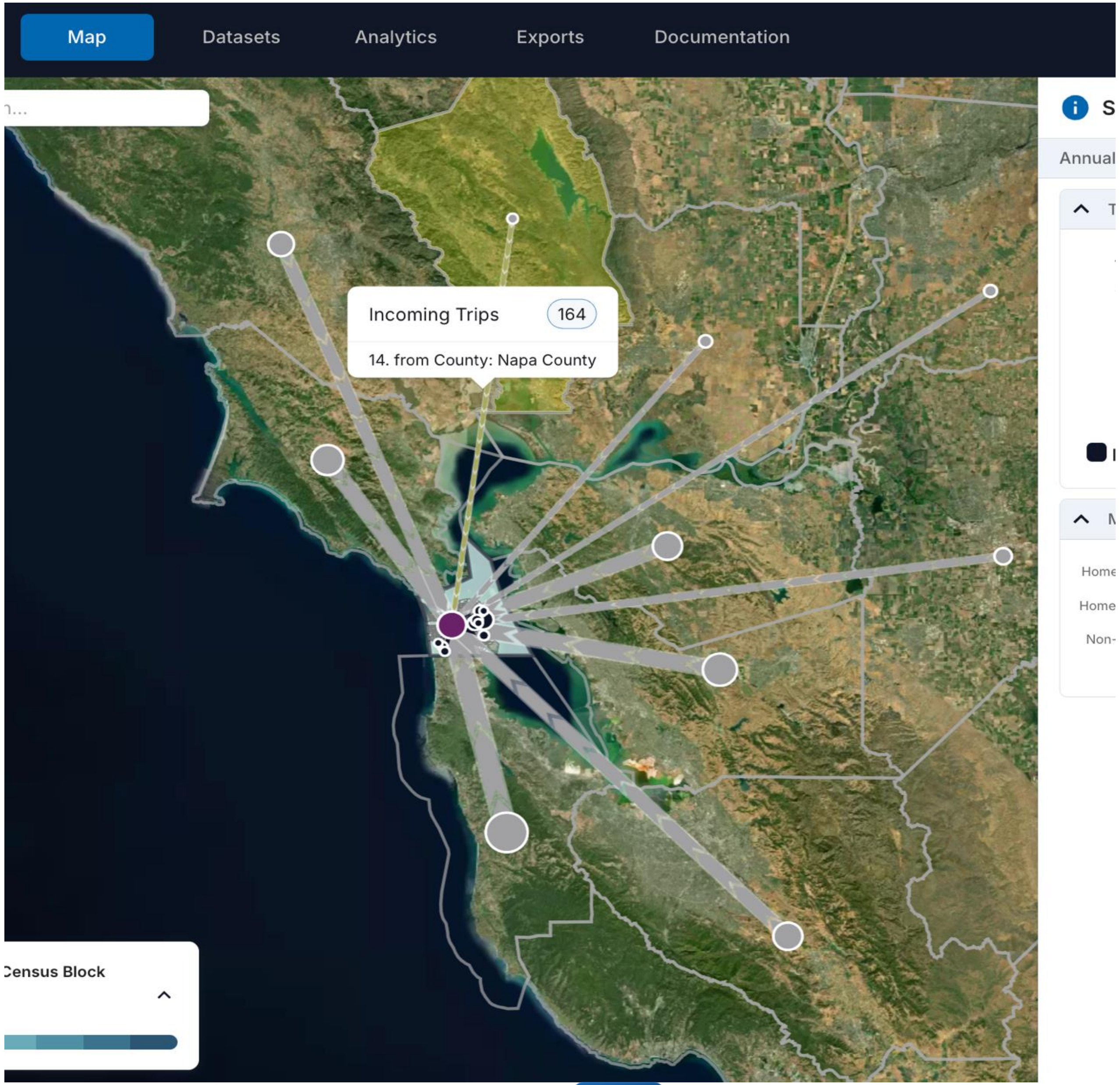
02 Core Features and Benefits

Origin Destination

Bring in your own data – TAZs, municipalities, special generators, etc.

- Average Annual ODs
- Census Block Group, Uber's H3 Grid, or any other structure
- Nationwide Data, Off-The-Shelf
- Quality Checked



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Roadway Volumes

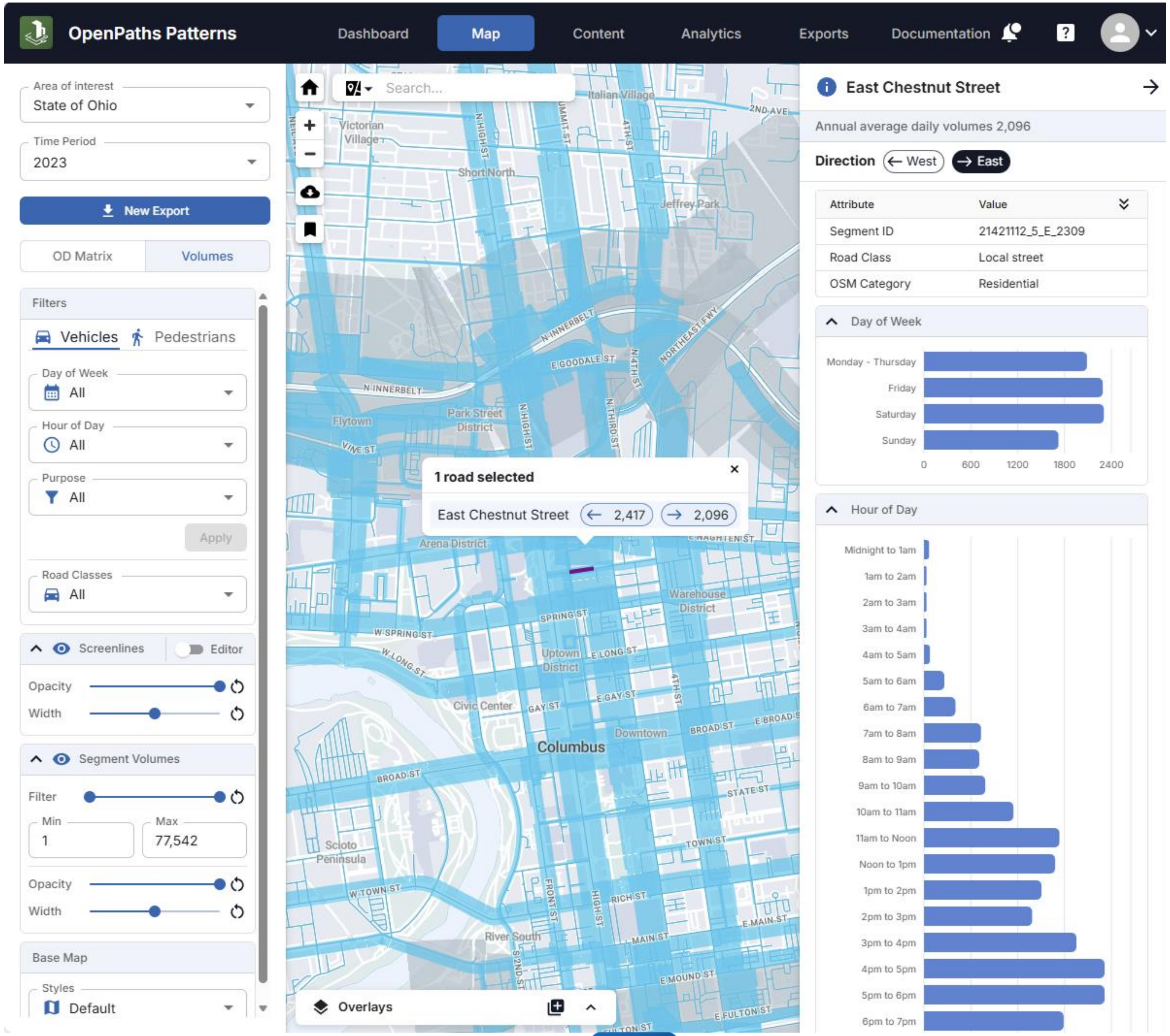
Customize map layers and attributes on the fly

Functional Class 1-7 Roadways Covered

Nationwide Data, Off-The-Shelf

Number of vehicles on all roads by:

- Direction
- Time of day
- Day of week
- Motivation of trip



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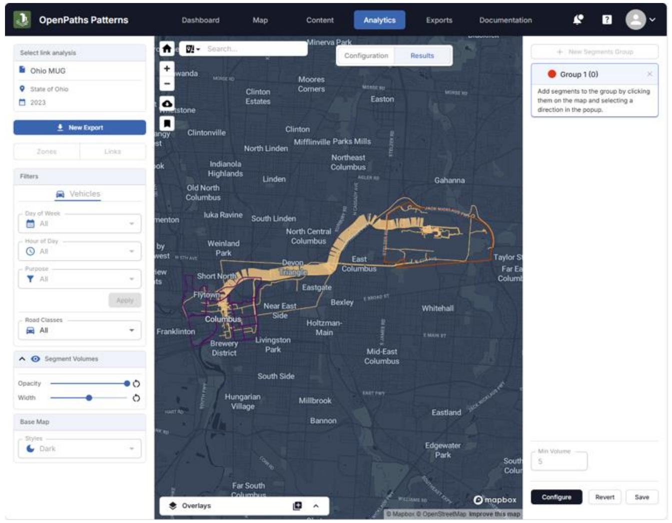
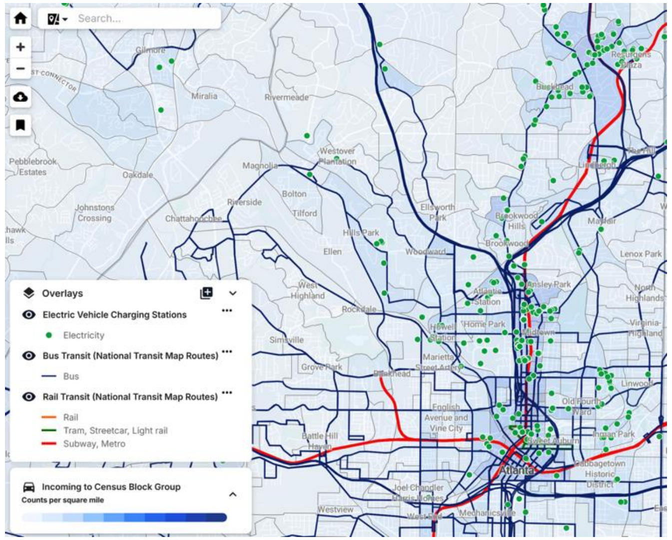
Analytics

Intuitive and powerful user-focused tools



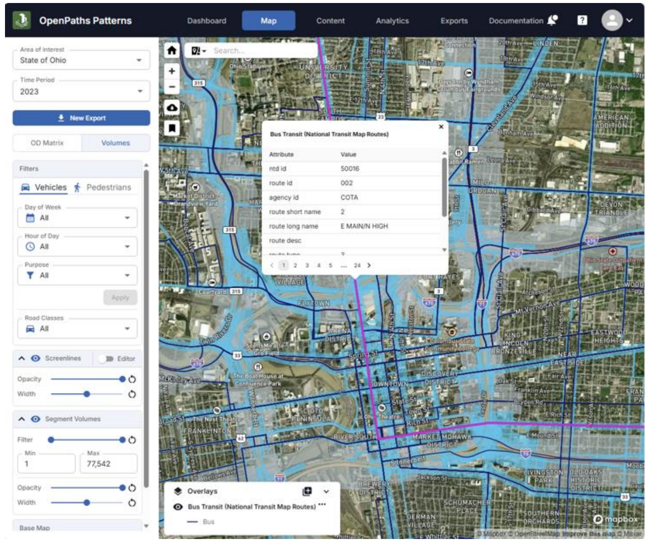
Select Link & Zone

Understand and quantify movements across a link, between two zones, to/from one zone, and much more. Enables geographic VMT analysis



Screenlines

Create and edit screenlines to quickly summarize travel flows through corridors, across entire regions, or in and out of activity centers. Upload your own screenlines to compare directly to your own results.



Map Overlays

Map Overlays enable public GIS feature layers to be display within OpenPaths Patterns. Available in any mapping view.

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Select Link & Zone

Select Link

Query volumes and travelsheds by a specific or selection of segments

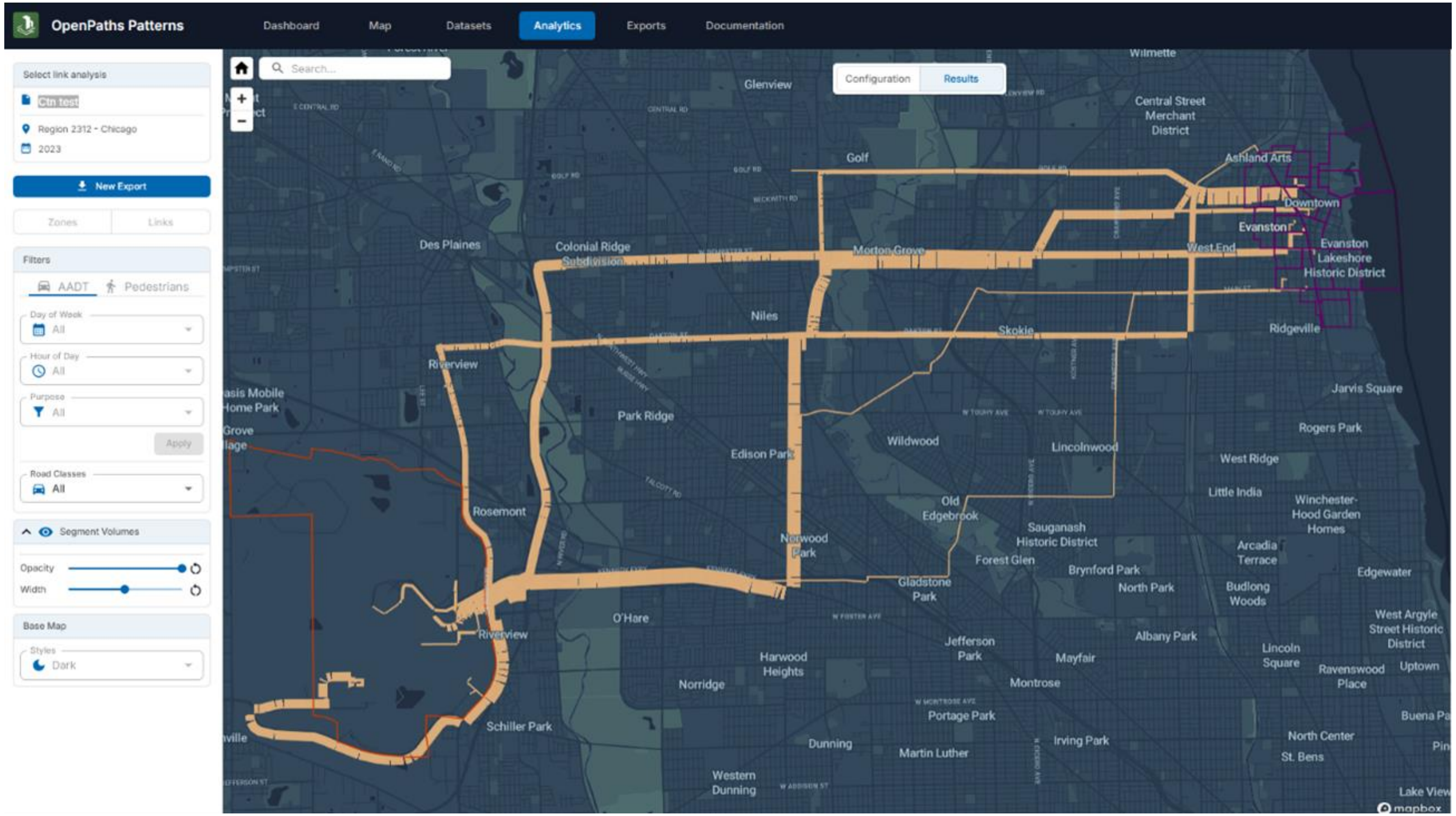
Select Zone

Query volumes and travelsheds by a specific or selection of geographies

Deep combinations

Utilize simple and/or logic to build specific queries for your analysis

**All information pulled from the select link & zone analysis is exportable for ingestion into your model!



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03 Real-World Case Studies

Accelerating DTA development
Sherburne County, MN

Purpose

Generate a versatile model for a small but growing county

Challenges

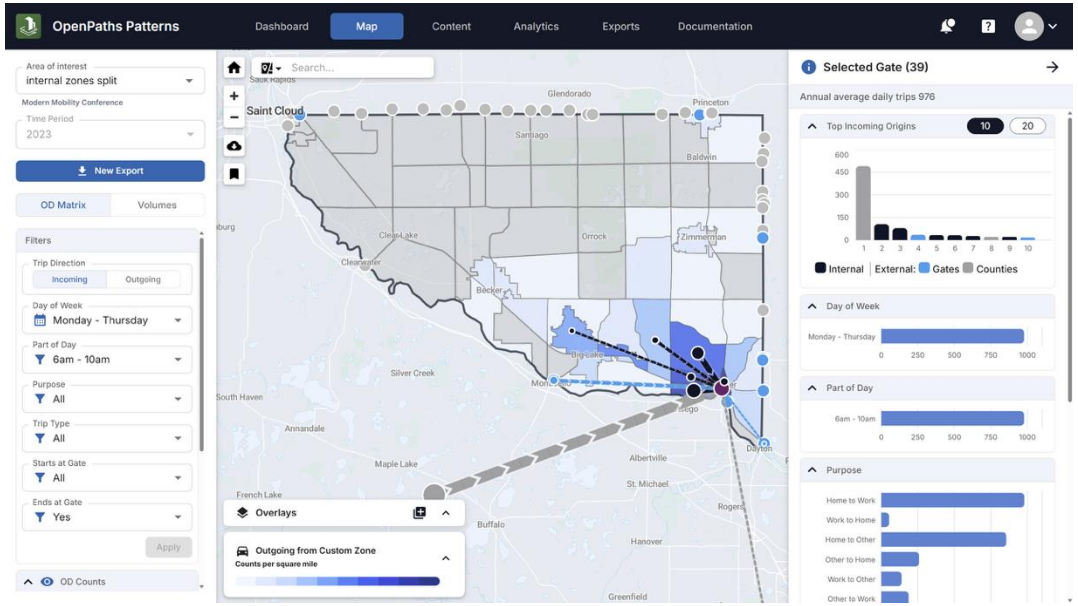
- Dynamic Traffic Assignment (DTA) is notoriously data heavy
- Community had little material to build model

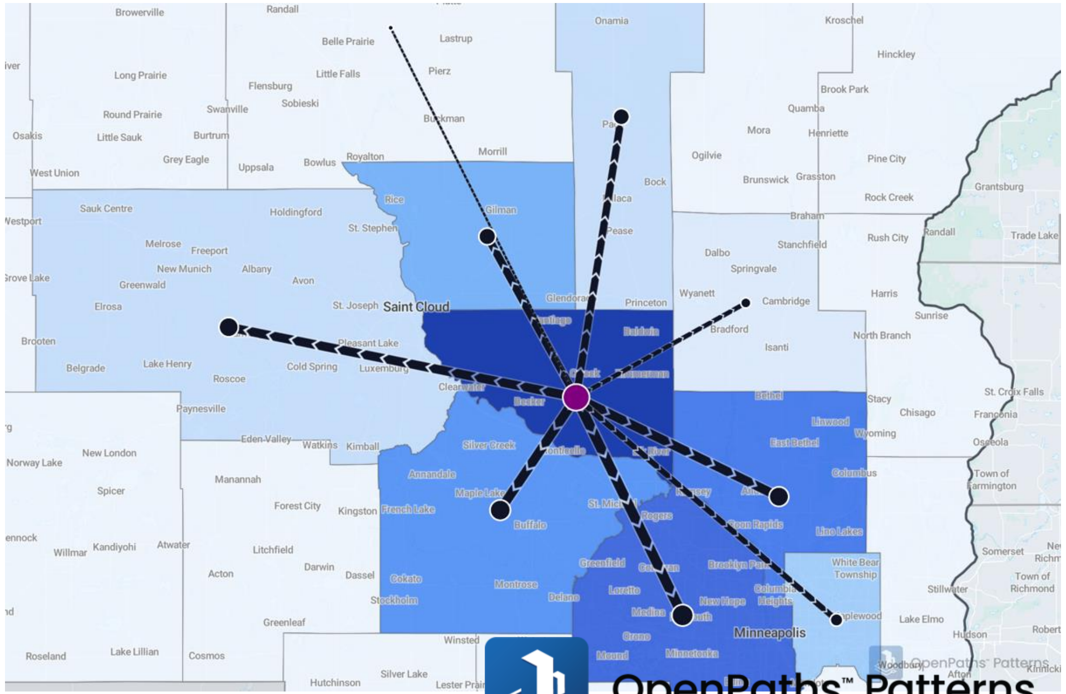
Project

Use OpenPaths Patterns to quickly create a model from scratch

Outcomes

1. Big data assisted in model development, with initial model running in 2 weeks
2. Cost-effective application for a small/medium sized community





Key Bridge Collapse
Baltimore, MD

Purpose

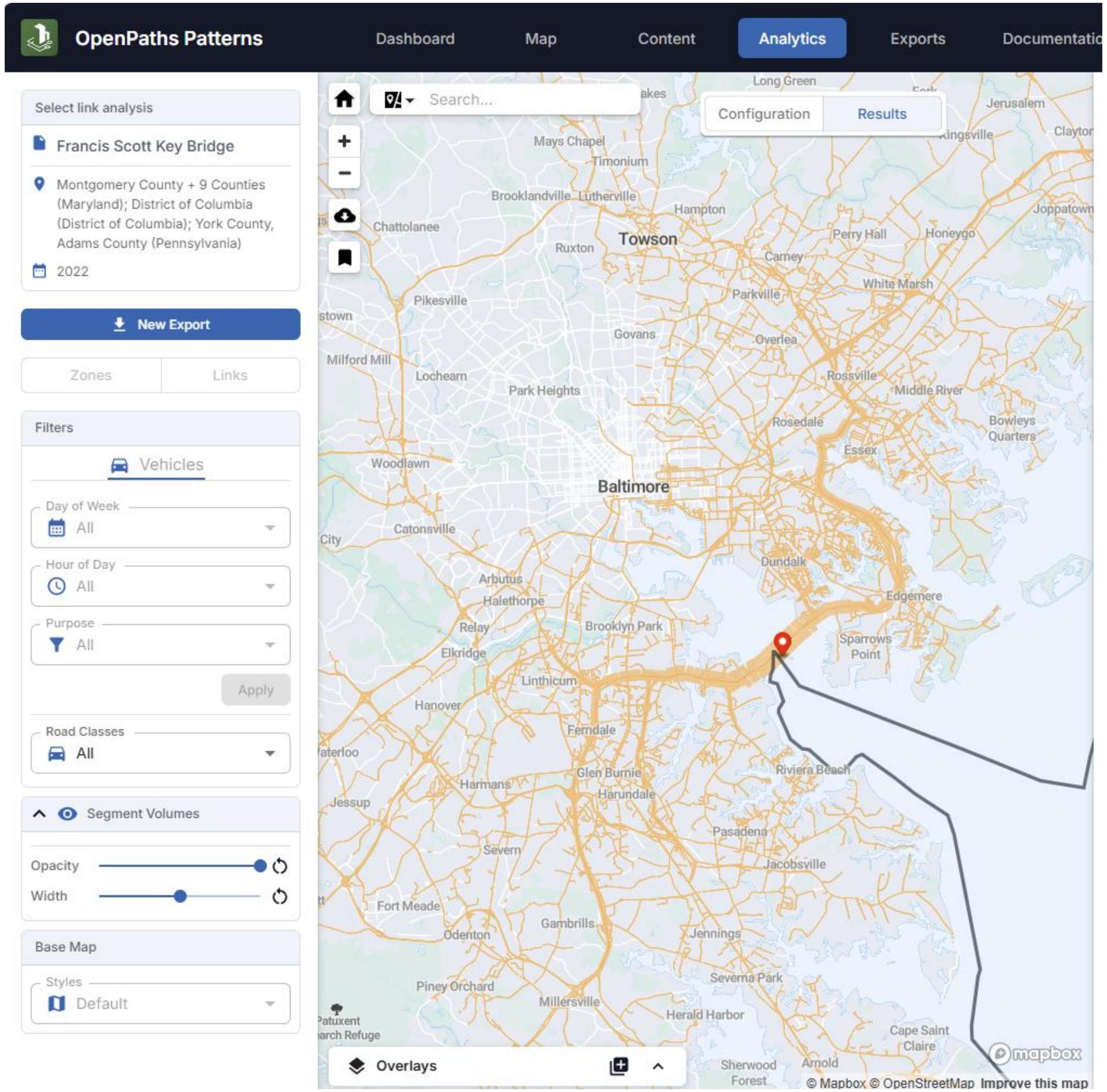
Quickly understand OD structure of trips on the bridge prior to collapse

Project

1. Use OpenPaths Patterns to restructure OD matrix to zip codes for entire region
2. Use OpenPaths Patterns to generate OD matrix of trips which utilized the Key Bridge (select link, export zones)

Outcomes

Within 48 hours the user was able to utilize the existing OD table to understand and plan for new routing challenges in the region



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04 Questions and Answers

How do I access OpenPaths Patterns?

OpenPaths Patterns is available for anyone with a Bentley account. A trial area (Sacramento) is set up to explore the features and capabilities.

www.bentley.com/software/openpaths-patterns

Go to URL above

This will bring you to the product page, which includes information about the product and access to the platform.

Create a Bentley account

If you do not have a Bentley account, create one via the product page.

Access the demo

If your company or organization has purchased a license, you will see it here. Otherwise, you will only see the demo area.

Add-on

We are here to support your needs, feel free to reach out any time at

openpaths-patterns@bentley.com

OpenPaths™ Patterns

How is OpenPaths Patterns offered?

We are excited to extend our most user centric approach to date. We have received positive feedback from users:

"This is a great opportunity for the modeling community, and a great offering for our work" – Regional Modeler

"With another vendor we needed to do 24 exports and manually compute an AADT – with OpenPaths Patterns we can compare an AADT within a minute" – Modeling Consultant

Statewide view-only access

View only access allows users to engage with the product at a low, fixed cost fee.

- Comprehensive data for all states
- All available data vintages and features
- Full software functionality, except for export

Export

Support modeling and local analysis with export, offered on a county-by-county basis. Contact us for pricing.

- All features in view-only, with addition of export
- Export enabled on a county-basis, rather than state basis

OpenPaths™ Patterns

Your OpenPaths Patterns Contacts

Ben Nault, ben.nault@Bentley.com

Katie Brinson, katie.Brinson@Bentley.com

Keith Hangland, keith.hangland@Bentley.com

Thank you, OTDMUG

Happy holidays and see you again soon at TRB 2026 in DC, Booth 1001

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